SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 wide notification number F-9-12/2001-U-3 of Government of India)

॥वसुधैव कुटुम्बकम्॥ Re-Accredited by NAAC with 'A++' Grade

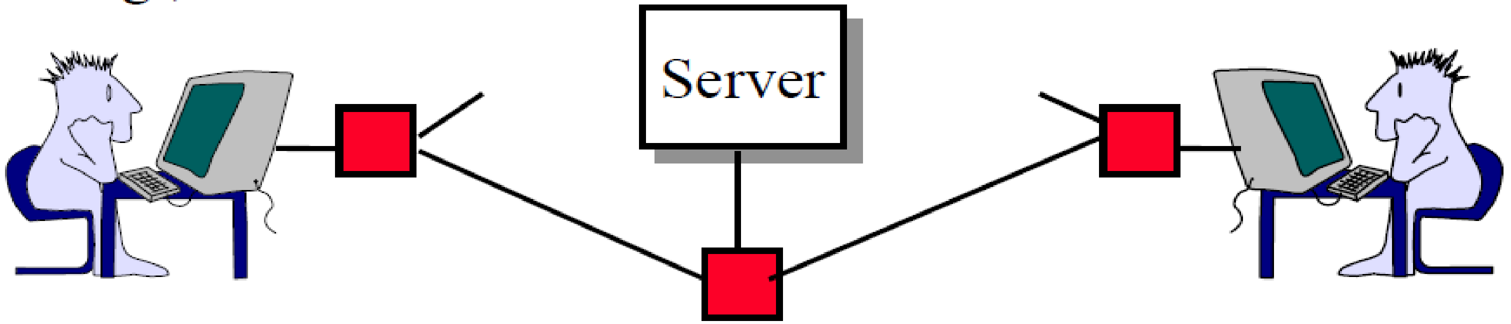
Celebrating 50 Years of Excellence

Computer Networks/0705210503

Prof. Soubhagya Ranjan Mallick
Assistant Professor (Selection Grade), CSE Dept.
SIT, Nagpur

What is a Network?

- **Network**: Enables data transfer among nodes
  - Generally heterogeneous nodes
  - More than <u>two</u> nodes
  - E.g., Your home or office network



- **Communication**: Two nodes.
  - Link level electrical issues.

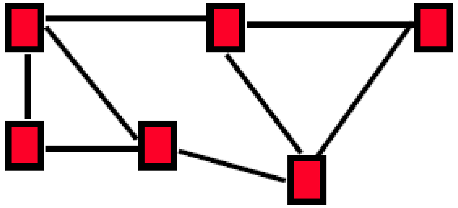
Key Concepts

- **End Systems**: Systems that are sinks or sources of data, e.g., Desktops, Laptops, Servers, Printers, Cell Phones, etc.
- **Intermediate Systems**: Systems that forward/switch data from one link to another, e.g., routers, switches
- **Hosts**: End Systems
- **Gateways**: Routers
- **Servers**: End Systems that provide service, e.g., print server, storage server, Mail server, etc.
- **Clients**: End systems that request service
- **Links**: Connect the systems.
  Characterized by transmission rate, propagation delay

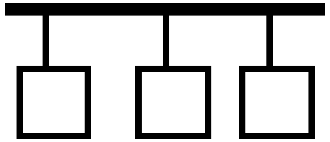
Types of Networks

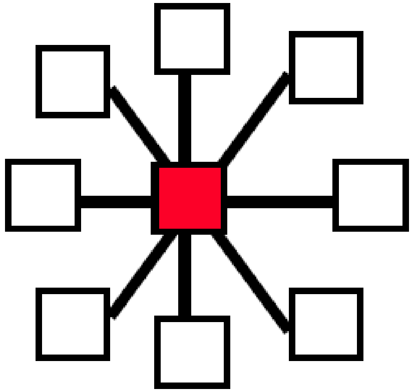
- **Point to point vs. Broadcast**

Point-to-Point



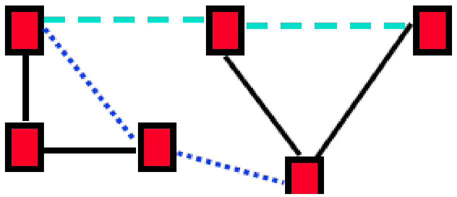
Bus

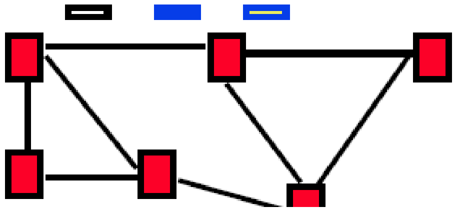




Star

- **Circuit switched vs. packet switched**
  - **Circuit:** A path (circuit) is setup before transmission. All bits follow the same path, e.g., Phone
  - **Packet:** Packets of bits are forwarded individually



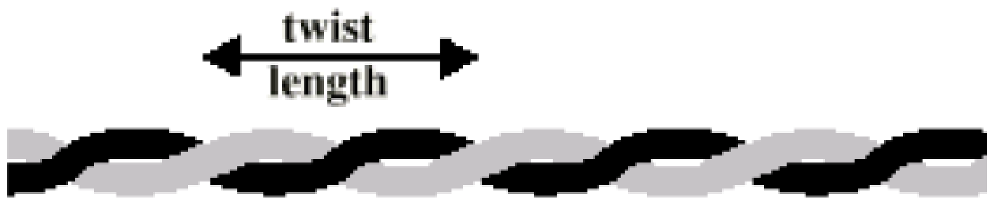


Transmission Media

- **Guided**:
  - Twisted Pair
  - Coaxial cable
  - Optical fiber
- **Unguided**:
  - Microwave
  - Satellite
  - Wireless

Twisted Pair (TP)

—Separately insulated
—Twisted together
—Often "bundled" into cables
—Usually installed in building during construction



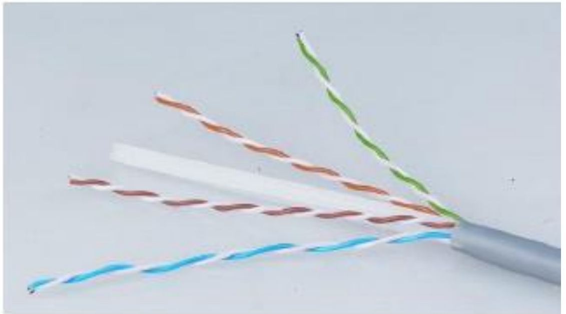
(a) Twisted pair

- Twists decrease the cross-talk
- Neighboring pairs have different twist length
- Most of telephone and network wiring in homes and offices is TP.

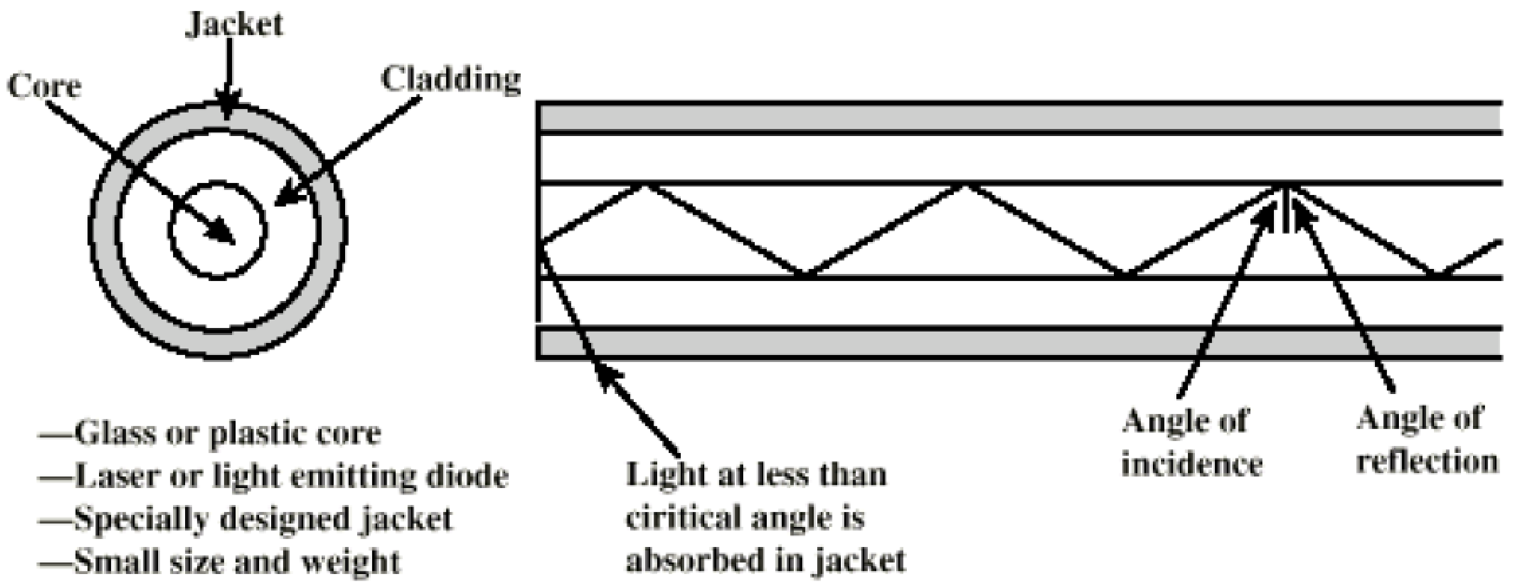
Shielded and Unshielded TP

- **Shielded Twisted Pair (STP)**
  - Metal braid or sheathing that reduces interference
  - More expensive
  - Harder to handle (thick, heavy)
  - Used in token rings

- **Unshielded Twisted Pair (UTP)**
  - Ordinary telephone wire
  - Cheap, Flexible
    $\Rightarrow$ Easiest to install
  - No shielding
    $\Rightarrow$ Suffers from external interference
  - Used in Telephone and Ethernet



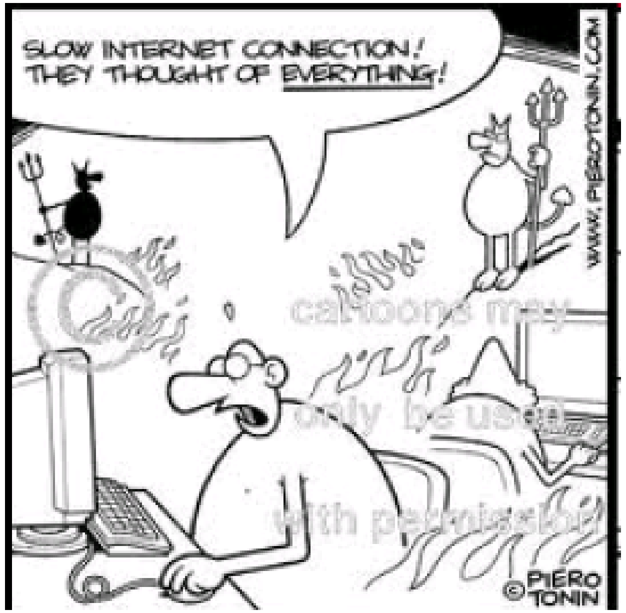
Optical Fiber



- A cylindrical mirror is formed by the cladding
- The light wave propagate by continuous reflection in the fiber
- Not affected by external interference $\Rightarrow$ low bit error rate
- Fiber is used in all long-haul or high-speed communication
- Infrared light is used in communication

Access Networks

1. DSL (Digital Subscriber Line)
2. Cable
3. Fiber-To-The-Home
4. Wi-Fi
5. LTE (Long Term Evolution)

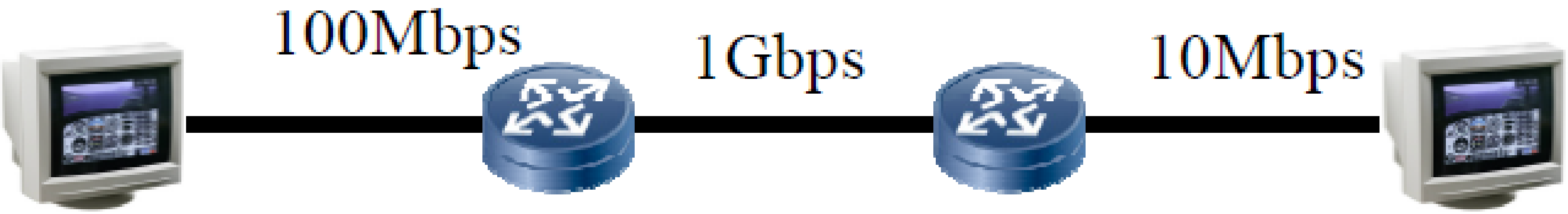


Network Performance Measures

- Delay
- Throughput
- Loss Rate

Throughput

- Measured in Bits/Sec
- Capacity: Nominal Throughput
- Throughput: Realistic
- Bottleneck determines the end-to-end throughput

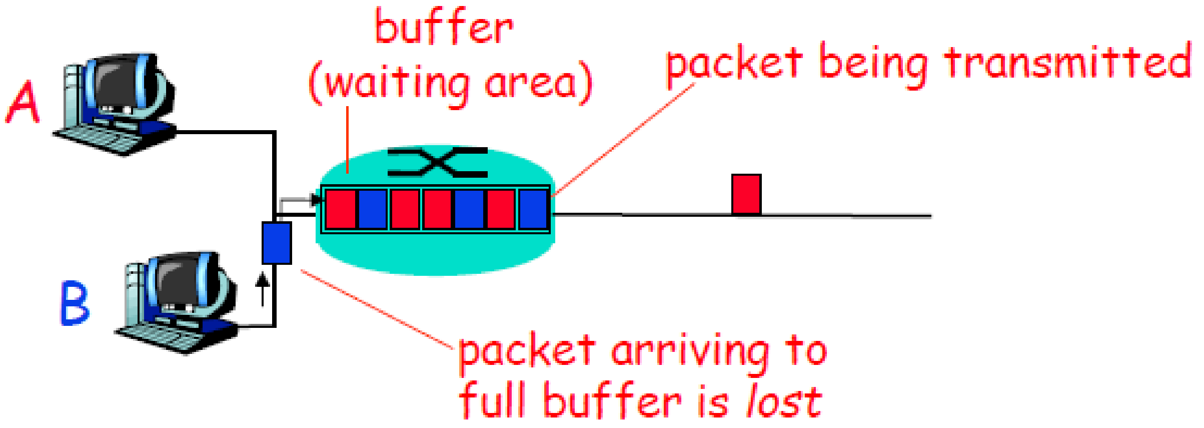


Net end-to-end capacity = 10 Mbps

Actual throughput will be less due to sharing and overhead.

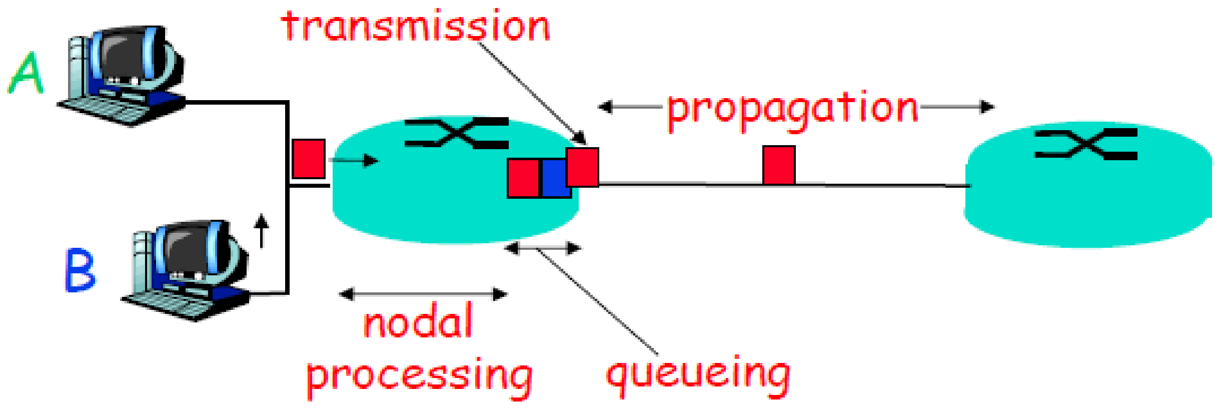
Loss Rate

- Queuing ⇒ Buffer overflow
- Bit Error Rate on the link
- Lost packets are retransmitted by the previous node or the source



Packet Switching Delay

1. **Processing Delay**: Check packets, decide where to send, etc.
2. **Queuing Delay**: Wait behind other packets
3. **Transmission Delay**: First-bit out to last-bit out on the wire = **Packet Length/bit rate**
4. **Propagation Delay**: Time for a bit to travel from in to out = Distance/speed of signal
5. **Speed of Signal**: **300 m/μs** light in vacuum, **200 m/μs** light in fiber, **250 m/μs** electricity in copper cables



Packet Switching Delay: Example

- 1500 Byte packets on 10 Mbps Ethernet, 1km segment
- Transmission Delay = $1500 \times 8 / 10 \times 10^6 = 1200\ \mu s = 1.2 ms$
- Propagation delay = $1000\ m / 2.5 \times 10^8 = 4\ \mu s$